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Palladium-Catalyzed Insertion of Allenes into Dithioacetals To Access 1,3-Dithioethers

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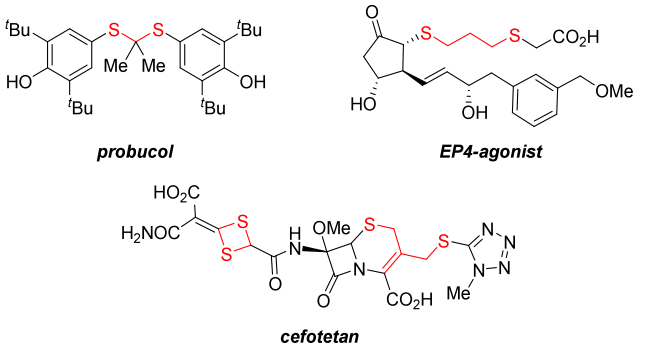
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ABSTRACT: Great efforts have been devoted to synthesizing dithioethers due to their unique pharmaceutical properties. While significant success has been achieved in the synthesis of 1,1-dithioethers and 1,2-dithioethers, a concise and efficient strategy for the direct construction of 1,3-dithioethers has not yet been established. Herein, we developed a palladium-catalyzed thiomethylthiolation of allenes with dithioacetals to access a wide range of 1,3-dithioethers through the simultaneous construction of one carbon−carbon bond and one carbon−sulfur bond. A dithioacetal-coordinated cationic palladium complex was isolated and characterized, which was considered the crucial intermediate for this palladium-catalyzed transformation.

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The carbon−sulfur linkage is one of the most fundamental connections omnipresent in numerous natural products, metabolites, and materials.[1−3] Especially, in the field of medicinal chemistry, sulfur-containing drugs were considered as the hallmarks of several antibiotic breakthroughs and continued to represent an important proportion of new U.S. Food and Drug Administration (FDA) approvals.[4] Among all of the bioactive organosulfur compounds, dithioethers were gaining increasing attention due to their unique pharmaceutical properties.[5] As shown in Scheme 1, probucol is a powerful antioxidant with modest lipid-lowering properties, which is of great clinical significance for coronary heart disease treatment.[6] Besides, cefotetan is a semisynthetic cephamycin antibiotic with a board spectrum of activity against Gram-negative bacteria.[7] As such, significant efforts have been dedicated to the synthesis of dithioethers, with the direct dithiolation of alkenes or alkynes being the most straightforward approach. However, in comparison to the elegant success for the synthesis of 1,1-dithioethers and 1,2-dithioethers,[8,9] the direct difunctionalization of alkenes to access 1,3-dithioethers has not been developed, which is mainly due to the lack of an appropriate dithiolation system. Considering the significant importance of 1,3-dithioethers in medicinal chemistry, developing an efficient method for their synthesis is of great value.

Scheme 1. Representative Dithioether Compounds in Bioactive Molecules



Dithioacetals are versatile synthons frequently used as the mask of carbonyl groups as well as valuable synthetic precursors for certain organosulfur compounds in transition metal catalysis.[10,11] The reactions involving insertion into the carbon−sulfur bond of dithioacetals provide a privileged platform for the rapid construction of functionalized dithioether compounds. Palladium-catalyzed thiol group migrations, which involve the formal insertion of unsaturated 2π units into the carbon−sulfur bond of dithioacetals, have already been developed (Scheme 2a).[12] However, these transformations were limited to only intramolecular processes. Recently, isolated intermolecular cases had also been illustrated with isocyanides or diazo compounds.[13] Nevertheless, the substrate scopes were still restricted to C1 sources, resulting in only 1,2-dithioether compounds rather than the desired 1,3-dithioethers.

Our group had previously demonstrated the feasibility of the intermolecular aminomethylamination of allenes with aminals to synthesize 1,3-diamines (Scheme 2b).[14] Building upon this, we became particularly interested in exploring the potential of using dithioacetals as thiomethylthiolation reagents to construct 1,3-dithioethers. As part of our design concept, we

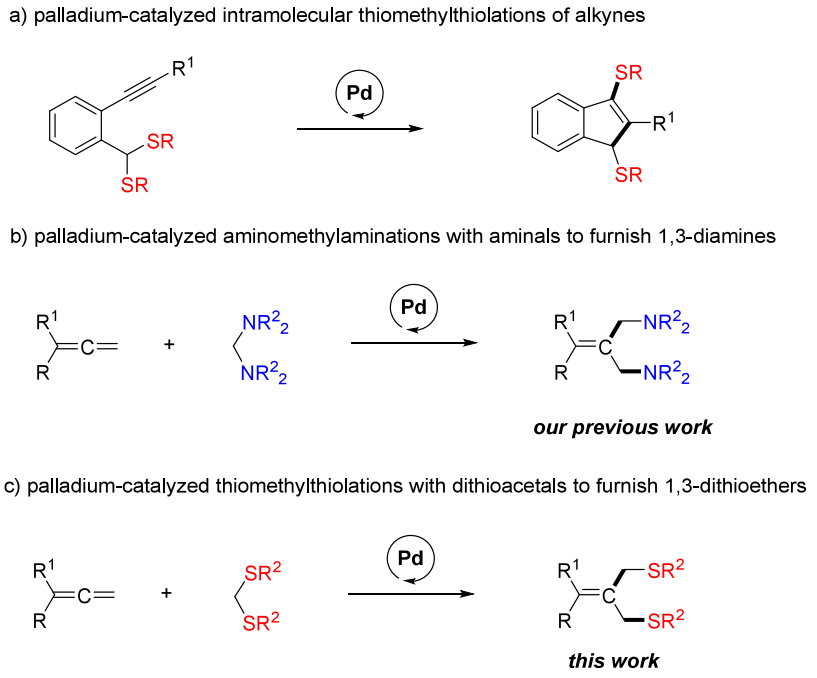
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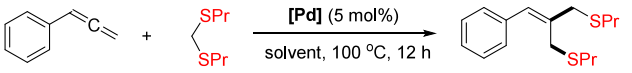
Scheme 2. Strategy for the Synthesis of 1,3-Dithioethers



aimed for the concurrent introduction of both the thiomethyl motif and the thiol motif of dithioacetals into allenes in a single step, similar to the aminomethylamination process using aminals. Herein, we successfully implemented this idea and developed a palladium-catalyzed insertion of allenes into dithioacetals to access 1,3-dithioether compounds (Scheme 2c). Distinguished from previous dithiolation processes, which solely depended upon the carbon−sulfur bond formation, this method allowed for the simultaneous construction of one carbon−carbon bond as well as one carbon−sulfur bond. It should also be noted that alkyl sulfur sources, which could hardly be tolerated in preceding dithiolation reactions, were well amendable to this protocol. This method offered an expedient solution for the rapid construction of a variety of structurally diverse 1,3-dithioethers, which can be used as valuable synthons and pharmacophores ubiquitously found in numerous natural products.

We embarked our investigations by treating propa-1,2-dienylbenzene **1a** and dithioacetal **2a** with the cationic palladium(II) complex in 1,2-dichloroethane (DCE) at 100 °C (Scheme 3).[15] To our delight, when [Pd(DPEPhos)-$(CH_3CN)_2$]$(OTf)_2$ was used as the catalyst, desired 1,3-dithioether **3a** was obtained in 34% yield (entry 1 in Scheme 3). Inspired by this result, a systematic screening of the counterion of the palladium complex was performed next, leading us to find that the counterion had a delicate influence on this transformation (entries 2−8 in Scheme 3). No reaction occurred when PF_6^-, OTs^-, or TFA^- served as the counterions, whereas ClO_4^-, NTf_2^-, and BF_4^- effectively promoted the transformation. Among all of the counterions, SbF_6^- was identified as optimal, affording desired 1,3-dithioether **3a** in 52% yield. Next, the ligand effect was also investigated, showing that bidentate phosphine ligands, such as BINAP, DPPF, XantPhos, and DPPB, all facilitated the reaction, albeit with lower yields compared to DPEPhos (entries 9−12 in Scheme 3). Additionally, variations of the solvents suggested that this transformation also proceeded smoothly in other chlorinated hydrocarbon solvents (entries 13−15 in Scheme 3). Finally, the optimal reaction conditions were then achieved by increasing the palladium catalyst loading

Scheme 3. Optimization of Reaction Conditions[a]



1a **2a** **3a**

entry	[Pd]	solvent	yield (%)
1	$[Pd(DPEPhos)(CH_3CN)_2](OTf)_2$	DCE	34
2	$[Pd(DPEPhos)(CH_3CN)_2](PF_6)_2$	DCE	trace
3	$[Pd(DPEPhos)(CH_3CN)_2](ClO_4)_2$	DCE	43
4	$[Pd(DPEPhos)(CH_3CN)_2](NTf_2)_2$	DCE	50
5	$[Pd(DPEPhos)(CH_3CN)_2](BF_4)_2$	DCE	42
6	$[Pd(DPEPhos)(CH_3CN)_2](OTs)_2$	DCE	trace
7	$[Pd(DPEPhos)(CH_3CN)_2](TFA)_2$	DCE	trace
8	$[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$	DCE	52
9	$[Pd(BINAP)(CH_3CN)_2](SbF_6)_2$	DCE	26
10	$[Pd(DPPF)(CH_3CN)_2](SbF_6)_2$	DCE	48
11	$[Pd(Xantphos)(CH_3CN)_2](SbF_6)_2$	DCE	50
12	$[Pd(DPPB)(CH_3CN)_2](SbF_6)_2$	DCE	34
13	$[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$	DCM	47
14	$[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$	$CHCl_3$	35
15	$[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$	PhCl	42
16	$[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$	toluene	12
17[b]	$[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$	DCE	74 (72)
18	$Pd(DPEPhos)Cl_2$	DCE	0

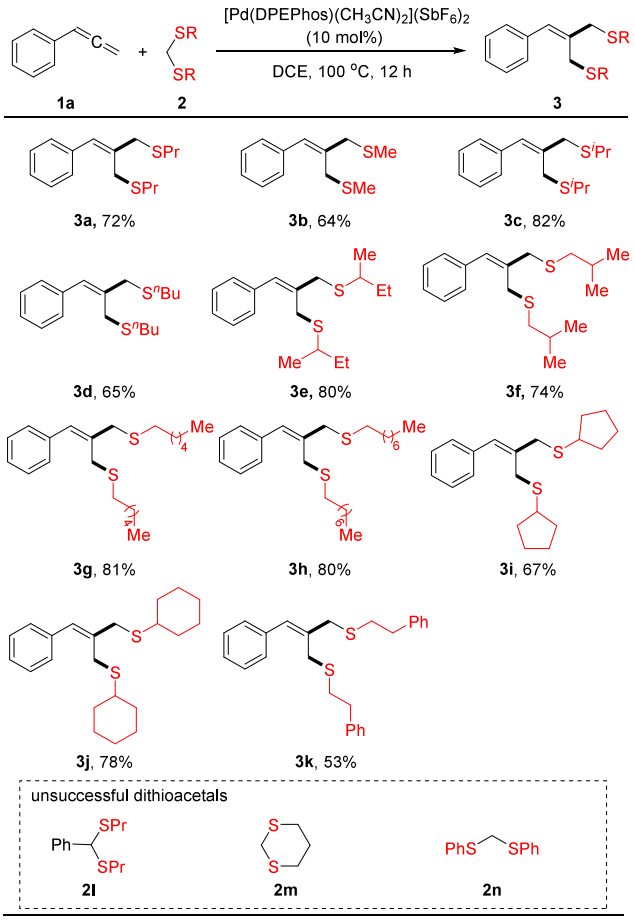
[a]Reaction conditions: allene **1a** (0.30 mmol), dithioacetal **2a** (0.20 mmol), [Pd] (5 mol %, 0.01 mmol), and solvent (1.5 mL), at 100 °C, for 12 h; yields were determined by gas chromatography (GC) using dodecane as an internal standard. [b][Pd] (10 mol %, 0.02 mmol); isolated yield was shown in parentheses.

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Org. Lett. 2025, 27, 2395−2399

to 10 mol %, resulting in target 1,3-dithioether **3a** in 72% yield (entry 17 in Scheme 3). The control experiment showed that no reaction took place when $Pd(DPEPhos)Cl_2$ was used as the palladium catalyst (entry 18 in Scheme 3), indicating that an appropriate cationic palladium(II) complex was necessary.

With the optimized reaction conditions in hand, we next evaluated the scope of this transformation. The substrate scope of dithioacetals was first investigated. As shown in Scheme 4,

Scheme 4. Substrate Scope of Dithioacetals[a]

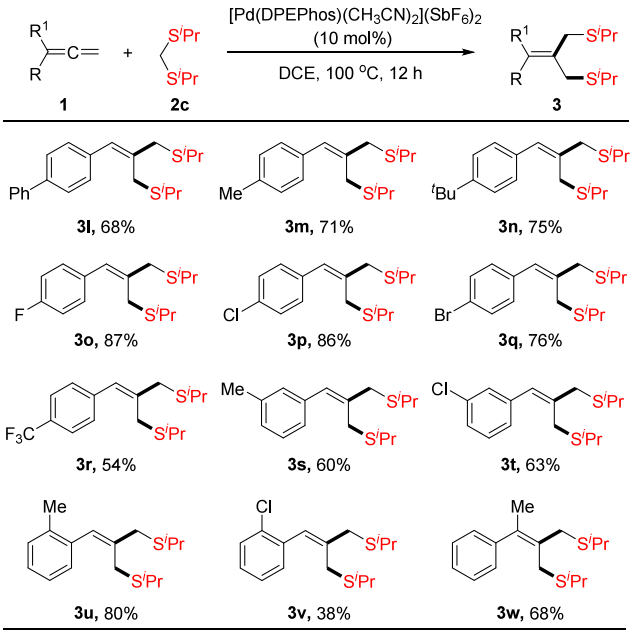


[a]Reaction conditions: allene **1a** (0.30 mmol), dithioacetal **2** (0.20 mmol), $[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$ (10 mol %, 0.02 mmol), and DCE (1.5 mL), at 100 °C, for 12 h.

two methylthio functionalities were simultaneously introduced into allene **1a** using 2,4-dithiapentane **2b** as dithioacetal, affording desired 1,3-dithioether **3b** in 64% yield. Additionally, dithioacetals prepared from structurally diverse alkyl thiols were investigated as well, furnishing the corresponding 1,3-dithioethers in 65−82% yields (**3c**−**3f**). Next, the insertion of allene **1a** into long-chain thiol-derived dithioacetals also proceeded smoothly, providing the corresponding 1,3-dithioethers in excellent yields (**3g** and **3h**). Furthermore, cyclopentylthiol- and cyclohexylthiol-derived dithioacetals were also well-tolerated in this catalytic system, giving the target 1,3-dithioethers **3i** and **3j** in 67 and 78% yields, respectively. Moreover, bis(phenethylthio)methane **2k** was also amenable to this system, providing 1,3-dithioether product **3k** in 53% yield. However, substituted dithioacetal (**2l**), cyclic dithioacetal (**2m**), and bis(phenylthio)methane (**2n**) did not yield the desired product in this system.

We next explored the scope of allenes using dithioacetal **2c** as the coupling partner. As illustrated in Scheme 5, arylallenes

Scheme 5. Substrate Scope of Allenes[a]

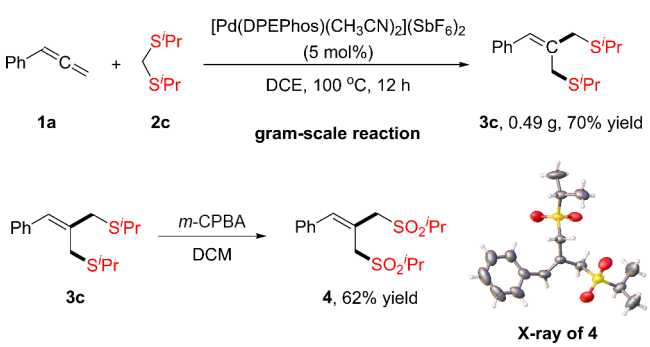


[a]Reaction conditions: allene **1** (0.30 mmol), dithioacetal **2c** (0.20 mmol), $[Pd(DPEPhos)(CH_3CN)_2](SbF_6)_2$ (10 mol %, 0.02 mmol), and DCE (1.5 mL), at 100 °C, for 12 h.

bearing all sorts of functional groups on the phenyl ring were examined, which revealed that both electron-donating groups, such as phenyl (**3l**) and alkyl (**3m**, **3n**, and **3s**), as well as electron-withdrawing groups, such as fluoro (**3o**), chloro (**3p** and **3t**), bromine (**3q**), and trifluoromethyl (**3r**), were all compatible with the reaction conditions. In addition, the presence of a methyl group at the *ortho* position of the aryl ring showed no significant influence on the reactivity of this transformation, whereas a bulkier chloro substituent evidently diminished the yield (**3u** versus **3v**). Furthermore, *gem*-disubstituted allene **1w** was also applicable to this system, affording corresponding 1,3-dithioether **3w** in 68% yield. Unfortunately, the alkyl-substituted allenes failed to yield any desired products in our system.

To demonstrate the synthetic potential of this novel palladium-catalyzed thiomethylthiolation, a large-scale synthetic procedure were conducted. As illustrated in Scheme 6, the reaction proceeded smoothly on a gram scale when 5.0 mol

Scheme 6. Synthetic Transformations

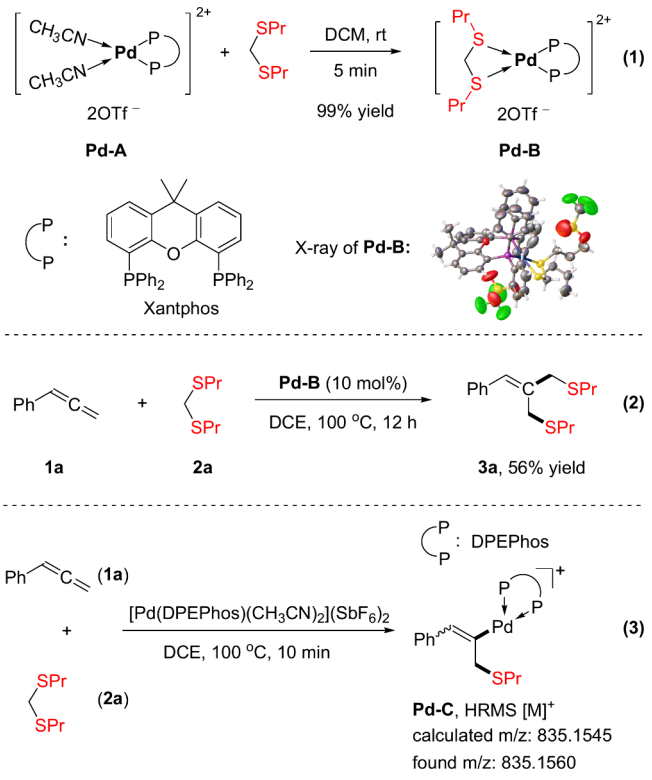


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% palladium catalyst was used, furnishing corresponding 1,3-dithioether **3c** in 70% yield. Additionally, oxidation of 1,3-dithioether **3c** with 3-chlorobenzoperoxoic acid (*m*-CPBA) took place successfully, affording 1,3-disulfone **4** in 62% yield. The solid structure of 1,3-disulfone **4** was characterized through X-ray crystallographic analysis.

To gain some insight into the mechanism of this transformation, a series of control experiments were next performed. As shown in Scheme 7, treating cationic palladium

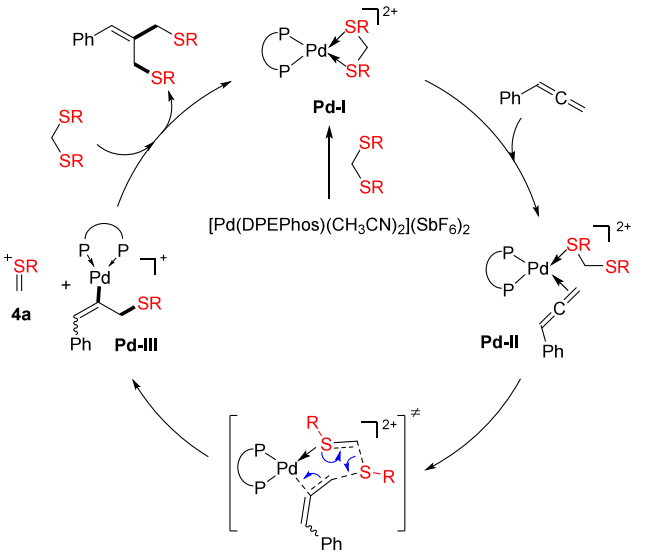
Scheme 7. Mechanistic Studies



complex **Pd−A** with 1.5 equiv of dithioacetal **2a** in dichloromethane (DCM) at room temperature rapidly produced dithioacetal-coordinated cationic palladium complex **Pd−B** (eq 1 in Scheme 7). The solid structure of **Pd−B** was unambiguously characterized by X-ray crystallographic analysis. Subsequently, the standard reaction proceeded smoothly in the presence of **Pd−B**, furnishing the desired 1,3-dithioether product **3a** in 56% yield (eq 2 in Scheme 7). These reactions collectively suggested that dithioacetal-coordinated cationic palladium complex **Pd−B** was most likely involved in this transformation. Furthermore, after 10 min of the standard reaction, we performed the high-resolution mass spectrometry (HRMS) analysis of the reaction mixture (eq 3 in Scheme 7), which detected that alkenyl palladium **Pd−C** was the main palladium species (see the Supporting Information for details).

On the basis of the aforementioned mechanistic studies and previous reports,[16] we proposed a plausible mechanism for the palladium-catalyzed thiomethylthiolation of allenes with dithioacetals (Scheme 8). Initially, ligand exchange took place to form dithioacetal-coordinated cationic palladium complex **Pd−I**, which was considered to be the resting state of this transformation. Upon the decoordination of one thioether motif of dithioacetal, the terminal carbon−carbon double bond of allene was able to coordinate to the cationic

Scheme 8. Plausible Reaction Mechanism



palladium complex to form **Pd−II**. Subsequently, nucleophilic attack of the decoordinated thioether motif to activated allene formed alkenyl palladium **Pd−III**, which was exactly detected by HRMS analysis. In the meantime, sulfonium species **4a** was formed via carbon−sulfur bond cleavage. Finally, migratory insertion of sulfonium **4a** into the C(sp^2)−Pd bond took place, yielding product 1,3-dithioether and regenerating cationic palladium complex **Pd−I** through dithioether exchange. The nucleophilic attack of the C(sp^2)−Pd bond to sulfonium **4a** also made it possible to complete the catalytic cycle.

In summary, we developed novel palladium-catalyzed thiomethylthiolations of allenes with dithioacetals, allowing for the rapid synthesis of a wide range of 1,3-dithioethers. A unique dithioacetal-coordinated cationic palladium complex was isolated and characterized, which was considered the crucial palladium intermediate for this transformation. Further application and the detailed mechanism of this transformation are currently ongoing in our laboratory and will be reported in due course.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00257.

Experimental procedures, characterization data, and copies of nuclear magnetic resonance (NMR) spectra for all novel compounds (PDF)

Accession Codes

Deposition numbers 2417175 and 2417177 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) (a) Devendar, P.; Yang, G.-F. Sulfur-Containing Agrochemicals. *Top. Curr. Chem.* **2017**, *375*, 82−126. (b) Wang, N.; Saidhareddy, P.; Jiang, X. Construction of sulfur-containing moieties in the total synthesis of natural products. *Nat. Prod. Rep.* **2020**, *37*, 246−275.

(2) (a) Cinar, M. E.; Ozturk, T. Thienothiophenes, dithienothiophenes, and thienoacenes: syntheses, oligomers, polymers, and properties. *Chem. Rev.* **2015**, *115*, 3036−3140. (b) Turkoglu, G.; Cinar, M. E.; Ozturk, T. Thiophene-Based Organic Semiconductors. *Top. Curr. Chem.* **2017**, *375*, 84−129.

(3) (a) Alegre-Cebollada, J.; Kosuri, P.; Rivas-Pardo, J. A.; Fernández, J. M. Direct observation of disulfide isomerization in a single protein. *Nat. Chem.* **2011**, *3*, 882−887. (b) Benavides, G. A.; Squadrito, G. L.; Mills, R. W.; Patel, H. D.; Isbell, T. S.; Patel, R. P.; Darley-Usmar, V. M.; Doeller, J. E.; Kraus, D. W. Hydrogen sulfide mediates the vasoactivity of garlic. *Proc. Natl. Acad. Sci. U. S. A.* **2007**, *104*, 17977−17982.

(4) (a) Teall, M.; Oakley, P.; Harrison, T.; Shaw, D.; Kay, E.; Elliott, J.; Gerhard, U.; Castro, J. L.; Shearman, M. R.; Ball, G.; Tsou, N. N. Aryl sulfones: a new class of γ-secretase inhibitors. *Bioorg. Med. Chem. Lett.* **2005**, *15*, 2685−2688. (b) Harrak, Y.; Casula, G.; Basset, J.; Rosell, G.; Plescia, S.; Raffa, D.; Cusimano, M. G.; Pouplana, R.; Pujol, M. D. Synthesis, anti-inflammatory activity, and in vitro antitumor effect of a novel class of cyclooxygenase inhibitors: 4-(aryloyl)phenyl methyl sulfones. *J. Med. Chem.* **2010**, *53*, 6560−6571. (c) Ilardi, E. A.; Vitaku, E.; Njardarson, J. T. Data-mining for sulfur and fluorine: an evaluation of pharmaceuticals to reveal opportunities for drug design and discovery. *J. Med. Chem.* **2014**, *57*, 2832−2842. (d) Feng, M.; Tang, B.; Liang, S. H.; Jiang, X. Sulfur containing scaffolds in drugs: Synthesis and application in Medicinal Chemistry. *Curr. Top. Med. Chem.* **2016**, *16*, 1200−1216.

(5) Mosberg, M. I.; Yeomans, L.; Anand, J. P.; Porter, V.; Sobczyk-Kojiro, K.; Traynor, J. R.; Jutkiewicz, E. M. Development of a Bioavailable μ Opioid Receptor (MOPr) Agonist, δ Opioid Receptor (DOPr) Antagonist Peptide That Evokes Antinociception without Development of Acute Tolerance. *J. Med. Chem.* **2014**, *57*, 3148−3153.

(6) Fukami, T.; Ishii, T.; Io, T.; Suzuki, N.; Suzuki, T.; Yamamoto, K.; Xu, J.; Ramamoorthy, A.; Tomono, K. Nanoparticle processing inthe solid state dramatically increases the cell membrane permeation ofa cholesterol-lowering drug, probucol. *Mol. Pharmaceutics* **2009**, *6*, 1029−1035.

(7) Iwanami, M.; Maeda, T.; Fujimoto, M.; Nagano, Y.; Nagano, N.; Yamazaki, A.; Shibanuma, T.; Tamazawa, K.; Yano, K. Synthesis of New Cephamycin Derivatives and a Novel Rearrangement between Isothiazolethioacetamides and 1, 3-Dithietanecarboxamides. *Chem. Pharm. Bull. Jpn.* **1980**, *28*, 2629−2636.

(8) Zhu, J.; Ye, Y.; Yan, Y.; Sun, J.; Huang, Y. Highly Regioselective Dichalcogenation of Alkenyl Sulfonium Salts to Access 1,1-Dichalcogenalkenes. *Org. Lett.* **2023**, *25*, 5324−5328.

(9) (a) Dong, B.; Shen, J.; Xie, L.-G. Recent Developments on 1,2-Difunctionalization and Hydrofunctionalization of Unactivated Alkenes and Alkynes Involving C-S Bond Formation. *Org. Chem. Front.* **2023**, *10*, 1322−1345. (b) Wei, C.; He, Y.; Wang, J.; Ye, X.; Wojtas, L.; Shi, X. Hexafluoroisopropanol-Promoted Disulfidation and Diselenation of Alkyne, Alkene, and Allene. *Org. Lett.* **2020**, *22*, 5462−5465. (c) Yu, G.; Ou, Y.; Chen, D.; Huang, Y.; Yan, Y.; Chen, Q. Air-Induced Disulfenylation of Alkenes: Facile Synthesis of Vicinal Dithioethers. *Synlett* **2020**, *31*, 83−86.

(10) (a) Luh, T. Y. New synthetic applications of the dithioacetal functionality. *Acc. Chem. Res.* **1991**, *24*, 257−263. (b) Luh, T.-Y. Recent advances on the synthetic applications of the dithioacetal functionality. *J. Organomet. Chem.* **2002**, *653*, 209−214. (c) Yan, J.; Pozzo, J.-L.; Hamze, A.; Provot, O. Recent Progress on the Mild Deprotection of Dithioketals, Dithioacetals, and Oxathiolanes. *Eur. J. Org. Chem.* **2022**, *2022*, e202101460.

(11) (a) Yus, M.; Nájera, C.; Foubelo, F. The role of 1,3-dithianes in natural product synthesis. *Tetrahedron* **2003**, *59*, 6147−6212. (b) Bur, S. K.; Padwa, A. The Pummerer Reaction: Methodology and Strategy for the Synthesis of Heterocyclic Compounds. *Chem. Rev.* **2004**, *104*, 2401−2432. (c) Smith, L. H. S.; Coote, S. C.; Sneddon, H. F.; Procter, D. J. Beyond the Pummerer Reaction: Recent Developments in Thionium Ion Chemistry. *Angew. Chem., Int. Ed.* **2010**, *49*, 5832−5844.

(12) Nakamura, I.; Bajracharya, G. B.; Wu, H.; Oishi, K.; Mizushima, Y.; Gridnev, I. D.; Yamamoto, Y. Catalytic Cyclization of o-Alkynylbenzaldehyde Acetals and Thioacetals. Unprecedented Activation of the Platinum Catalyst by Olefins. Scope and Mechanism of the Reaction. *J. Am. Chem. Soc.* **2004**, *126*, 15423−15430.

(13) (a) Tobisu, M.; Ito, S.; Kitajima, A.; Chatani, N. $GaCl_3$- and $TiCl_4$-Catalyzed Insertion of Isocyanides into a C-S Bond of Dithioacetals. *Org. Lett.* **2008**, *10*, 5223−5225. (b) Raskildina, G. Z.; Zlotsky, S. S.; Sultanova, R. M. Diazo Compounds in the Synthesis of O- and S-Containing Macroheterocycles. *Macroheterocycles* **2018**, *11*, 166−172.

(14) Hu, J.; Xie, Y.; Huang, H. Palladium-Catalyzed Insertion of an Allene into an Aminal: Aminomethylamination of Allenes by C−N Bond Activation. *Angew. Chem., Int. Ed.* **2014**, *53*, 7272−7275.

(15) Qin, G.; Li, L.; Li, J.; Huang, H. Palladium-Catalyzed FormalInsertion of Carbenoids into Aminals via C−N Bond Activation. *J. Am. Chem. Soc.* **2015**, *137*, 12490−12493.

(16) Hisano, N.; Kamei, Y.; Kansaku, Y.; Yamanaka, M.; Mori, K. Construction of 1,3-Dithio-Substituted Tetralins by [1,5]-Alkylthio Group Transfer Mediated Skeletal Rearrangement. *Org. Lett.* **2018**, *20*, 4223−4226.